

Mohamed Ahmed Abd El Salam El Seuofy

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Dear Sir,

I am pleased to introduce myself as a mechanical engineer and researcher with more than 15 years of professional experience in the oil and petrochemical industry, combined with a strong academic and research background in energy systems optimization, heat transfer, decarbonization, and hydrogen technologies.

My academic journey began with a Master's degree in mechanical engineering from Alexandria University, where my thesis, "*An Investigation of Heat Transfer in a Polymer Reactor and a Case Study*," focused on optimizing heat transfer in polymerization reactors to improve efficiency and reduce energy consumption.

This research provided practical industrial solutions and formed the basis for my first international publication in Thermal Science Journal, titled "A Simulation Study of Heat Transfer in Polymerization Reactors."

Currently, my Ph.D. research focuses on cryogenic hydrogen and green hydrogen systems, with emphasis on innovative approaches for hydrogen liquefaction using advanced thermodynamic cycles.

My recent research includes:

- *Energy, Exergy, Economic, and Environmental Assessment of a Transcritical Triple-Cascade Mixed Refrigerant Cycle for Hydrogen Liquefaction — Cryogenics Journal.*
- *Comparative Study of Advanced Hydrogen Liquefaction Using Triple Cascade Mixed Refrigerant Cycles — Scientific Reports, Springer Nature.*

These studies combine advanced simulation, exergy analysis, economic evaluation, and environmental assessment to develop hydrogen solutions that support both technical performance and global decarbonization objectives.

Alongside my academic work, I serve as Director of Energy Efficiency and Mechanical Equipment Follow-up at Alexandria Mineral Oils Company (AMOC). In this role, I lead energy optimization initiatives, mechanical equipment performance improvement, energy audits, and decarbonization activities. I have contributed to implementing ISO 50001, ISO 14064, and ISO 14067, improving energy performance, supporting GHG reporting, and reducing environmental impact.

My core expertise includes:

- Energy and exergy optimization of complex thermodynamic systems.
- Hydrogen storage and liquefaction technologies with advanced refrigeration cycle integration.
- Process simulation and thermal analysis using Aspen HYSYS and advanced modeling tools.
- Sustainability and ISO standards implementation for energy and carbon management.
- Development of digital platforms for ISO 14064 GHG calculation and reporting.

I am passionate about bridging scientific research with industrial applications, contributing to sustainable energy systems, and supporting practical hydrogen and decarbonization solutions for the energy sector.

Thank you for taking the time to review my profile. I would be honored to connect, collaborate, and contribute to knowledge-sharing initiatives that advance clean energy and hydrogen technologies.

Sincerely,

Dr. Mohamed Abd El Salam El Seuofy



MOHAMED AHMED ABD EL SALAM EL SEUOFY

Profile

I am a results-driven mechanical engineer, energy efficiency leader, and hydrogen researcher with over 15 years of experience in the oil and petrochemical industries. My career has focused on transforming engineering knowledge into practical improvements that enhance equipment reliability, reduce energy consumption, minimize downtime, and deliver measurable operational and financial value. I specialize in energy efficiency, mechanical equipment reliability, rotating equipment optimization, troubleshooting, and performance improvement. As a Certified Energy Manager (CEM), Certified Energy Auditor (CEA), and ISO 50001 Lead Auditor, I have hands-on experience in implementing ISO 50001, ISO 14064, and ISO 14067 systems, supporting energy management, greenhouse gas reduction, and carbon footprint reporting. I hold a Master's degree and Ph.D. in Mechanical Engineering, with a research focus on energy efficiency, cryogenic hydrogen, green hydrogen systems, and advanced hydrogen liquefaction technologies. My work bridges academic research and industrial application, aiming to support practical decarbonization solutions and the clean-energy transition. In addition to my technical and research background, I have strong teaching, mentoring, and training experience, helping engineers and industrial teams build capabilities, improve performance, and adopt a culture of continuous improvement and sustainability.

Experience

Managing Director of Energy Efficiency and Mechanical Equipment Follow-up

Alexandria Mineral Oils Company_ (AMOC)

November 2023 – Present

- Managed the performance of mechanical equipment, implementing maintenance and operational best practices that improved reliability and minimized energy consumption.
- Led company-wide energy optimization projects, achieving a 3.5% reduction in energy consumption across critical heat systems and electrical operations.
- Conducted training programs for engineers and staff to enhance technical competencies in energy and equipment management.
- Led energy audit assessments with a focus on operational performance, maintenance efficiency, and energy optimization.
- Increased LPG production by 20% through flare recovery.
- Implemented systems for ISO 50001, 14064, and 14067 to integrate quality management with maintenance, energy efficiency, and environmental sustainability.
- Generate and design platform for GHG calculation ISO 14064 and reporting.

Section Head of Rotary Equipment Maintenance and Energy Efficiency

Egyptian Petrochemicals Company _EPC

- Led the maintenance of 1000+ rotating equipment, ensuring strict adherence to standard operating procedures, and reducing downtime.

National of Country:
EGYPT

Current Job Title:
Section Head Of Energy
Efficiency

**Number of Years with
Companies:**
13

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- Oversee maintenance work progress in adherence to standard procedures.
- Supervised over 70 labor force of engineers and technicians in three different sections including reactors, compressors, and blowers.
- Identified and implemented energy-saving opportunities with the energy team, achieving a 12% reduction in energy consumption for the whole company.
- Proposed and implemented changes in centrifugal compressor equipment with the energy team to achieve annual power savings of 815 MW.
- Investigated deviations in rotating equipment processes, identified root causes, and recommended corrective actions, minimizing unscheduled downtime.
- Collaborated with inspection and reliability engineering teams to optimize reliability-centered maintenance, enhancing plant availability.

Site Engineer

Alexandria Petroleum Maintenance – Petromaint

July 2011 –
April 2012

- Daily management of the site and precise censorship on-site labor force.
- Maintaining the work is performed as per specifications and standard quality.
- Prepare the reports required and supervise the project schedule.

Education

7/ 2011	Bachelor of Mechanical Engineering, Alexandria University, Faculty of Engineering, 2011.
1/ 2018	Master of Science in Mechanical Engineering, Alexandria University, Faculty of Engineering, 2018.
1/ 2026	PhD in Mechanical Engineering Alexandria University, Faculty of Engineering, Alexandria University, Faculty of Engineering, 2026.

Skills

- Mechanical and Rotating Equipment Maintenance
- Hydrogen energy production and liquification
- 4E- Energy Exergy Economic Environment based experience studies
- HYSYS simulation of hydrogen plants and optimization
- Energy Management and Optimization (ISO 50001) – efficiency improvement & performance tracking
- Certified Energy Manager (CEM) & Certified Energy Auditor (CEA)
- Exceptional Problem-Solving & Analytical Skills
- Python coding to design platform for GHG calculation ISO 14064 and reporting.
- Process Simulation and System Optimization (data-driven decision support)
- Reliability-Centered Maintenance (RCM) and Maintenance Strategy Development
- Root Cause Analysis (RCA) and Structured Problem Solving for failures and performance losses
- Project Management for complex, cross-functional initiatives (scope, time, cost, quality)
- Leadership and Team Mentoring – multidisciplinary coordination and collaboration across departments
- Technical Reporting & Stakeholder Communication.
- Continuous Improvement & Proactive Mindset – measurable results, innovation, and optimization culture
- Technical Training and Capacity Building (coaching teams, knowledge transfer)

Teaching, Mentoring, And Coaching Experience

- **Mechanical Engineering Principles**

Dates: June 2018 - August 2019, 2020 - Location: **Egyptian Petrochemical Company (EPC)**

Key Activities: Conducted training sessions on mechanical engineering principles and best practices.

- **Energy Efficiency Techniques**

Dates: September 2021 - May 2023 - Location: **(EPC) and Alexandria Mineral Oils Company (AMOC)**

Key Activities: Provided mentoring on energy efficiency techniques and applications.

- **Advanced Rotating Equipment Maintenance**

Dates: August 2018, July 2019, 2020, 2021, 2022

Role: Instructor at **EPC Training Center**

Key Activities: Offered a comprehensive course on advanced techniques for maintaining and troubleshooting rotating equipment.

- **Energy Management System ISO 50001**

Dates: May 2023 - February 2024, May 2024, July 2024, December 2024

June 2025, July 2025, Sep 2025 , Feb 2026 , May 2026

Role: Trainer and qualification for various companies Location :

Egyptian Petrochemical Company (EPC)

Alexandria Mineral Oils Company (AMOC)

Assiut Oil Refining Company (ASORC)

Cairo Oil Refining Company (CORC)

Western Desert Operating Petroleum Co. (WEPCO)

Key Activities: Led training in energy management systems and ISO 50001 implementation

- **Energy Auditor ISO 50002**

Dates: June 2026 – Location: **Yemen Mobile**

Dates: May 2025 – Location: **KSA - TEKNOKEYS training center**

Training Certification

Technical Courses	Certificated by
Certified Energy Manager (CEM)	Association of Energy Engineers
Certified Energy Auditor – (CEA)	Association of Energy Engineers
ISO 50001 Lead Auditor	TÜV NORD
ISO 14067 -14064-1	TUV – Austria
ISO 14064-1 Lead Auditor	National Quality Institute (NQI)
Energy Saving Opportunity in Oil and Gas	Egyptian Petrochemicals Holding Company

Thermal & Solar Energy Efficiency	UNIDO - Industrial Efficiency Program
Chiller System Optimization	UNIDO - Industrial Efficiency Program
Motors, Compressed Air, and Pump System Optimization	UNIDO - Industrial Efficiency Program
Advanced Power Station Design and problem-solution	Alexandria University - Faculty of Engineering EPROM
Technical And Economic Feasibility Study	(OGS)- Oil and Gas Skills Company
Rotating Equipment Basics Design and Maintenance	Egyptian Petrochemical Company
Vibration Analysis	Alexandria University - Faculty of Science- Alexandria Egypt Egyptian Petrochemical Company
30-Hour Training - OSHA	Egyptian Petrochemical Company
PMP, Project Management Professional	TUV – Austria
Six Sigma and Risk Assessment Project	TUV – Austria
API 570	Egyptian Petrochemical Company
API 617	(OGS)- Oil and Gas Skills Company
Rotating Equipment Maintenance	NEBOSH
Petroleum Industry (Basic Exploration, Drilling, Reservoir, and Oil & Gas Field Processing)	PETROSAFE
Process Safety Management (PSM)	Arab academy for information and technology
HAZOP	PETROGLOB
Key Performance Indicators	OAPEC- Organization of Arab Petroleum Exporting Countries Arab Academy for Science, Technology
Flange Management	American Chamber
Carbon Footprint	American Chamber
Leadership and change management	Egyptian Petrochemical Company
Strategic management	Egyptian Petrochemical Holding Company
Presentation and communication skills	ESLSCA University
Human resources	
Interpersonal communication skills	
TOT Training of trainers	

Professional Affiliations & Activities

- 6/ 2022 **Energy Efficiency Advisor**, national energy efficiency program, Association of Energy Engineers, Certified Energy Manager, Auditor - contributed to national initiatives for promoting energy efficiency in industrial sectors.
- 1/ 2023 **Top 30 Energy Efficiency Middle Management**: In Energy Efficiency Committee in Egyptian Petroleum Ministry.

Membership:

Society of Petroleum Engineers (SPE)
Occupation Safety & Health Administration (OSHA) Association of Energy Engineers (AEE) Honors and Awards

- **Energy Efficiency Ideas Award**, Alexandria Mineral Oils Company, 2024 Awarded for innovative approaches to improving energy efficiency in the sector.

Publications

Master of Science: **“A Simulation Study of Heat Transfer in Polymerization Reactors”** Mechanical Engineering Department, Faculty of Engineering, Alexandria University, Alexandria, Egypt Original scientific paper- <https://doi.org/10.2298/TSCI170914009E>.

PhD: **“Comparative study of advanced hydrogen liquefaction using triple cascade mixed refrigerant cycles with integrated energy exergy economic and environmental analysis”** - Scientific Reports
Published - DOI: <https://www.doi.org/10.1038/s41598-025-14258-8>

“Energy, exergy, economic, and environmental assessment of a transcritical triple-cascade mixed refrigerant cycle for hydrogen liquefaction”- Cryogenics, Published,
<https://doi.org/10.1016/j.cryogenics.2025.104254>

Knowledge Of Software

- Auto cad 2D drafting and 3D modeling
- Aspen HYSYS, Petro-SIM and PYTHON

Design Platform

ISO 14064 Carbon Footprint Calculator and Reporting Platform:
“ <https://mohamed-elseuofy.github.io/ISO14064-ABD-ELSALM3/>”

language

Arabic: Mother Tongue.
English: Fluent in writing and speaking | IELTS: Academic Certificate.

Conference Speaker

- **EGYPT ENERGY SHOW - EGPES** | 2019-2020
Presented on "energy efficiency in polymer reactors applications"
- **Solar & Storage Live Egypt** | 2024
Speaker on "solar energy systems and carbon capture"
- **ADIPEC 2024** Exhibition & Conference | 2024
Delivered insights on "Hydrogen storage align with AIG Atlas Copco & Sustainability"